

Module 13 : (Full) Factorial randomized experiments

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Design of Experiments - Stat 140

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Factorial randomized experiments (s^m factorial design)

- Experiments that involve the simultaneously study of multiple m treatment factors, each at s levels.
- Why varying multiple factors at a time ?
- Factor : variable that can be modified (i.e., intervention).
- Real-world setting
- Often very difficult
- s^m treatment combinations

General setting

- A factorial experiment that involves $s_1, s_2, \dots, s_K$ factors with $m_1, m_2, \dots, m_K$ levels respectively is called a $s_1^{m_1}, s_2^{m_2}, \dots, s_K^{m_K}$ mixed-level factorial design.
- Today, we will discuss two-level full factorial (i.e., 2^K) design.
- Full : all the 2^K treatment combinations are used.
- We denote the K factors by $F_1, F_2, \dots, F_K$ and the two levels of each factor by -1 and $+1$.
- Possible in engineering and industrial settings, less so in social and biomedical experiments.

Lexicographic order of treatment combinations : example with two factors, each having two levels

Treatment	F_1	F_2	Unit POs
$(-1,-1)$	-1	-1	$Y_i(-1,-1)$
$(-1,+1)$	-1	+1	$Y_i(-1,+1)$
$(+1,-1)$	+1	-1	$Y_i(+1,-1)$
$(+1,+1)$	+1	+1	$Y_i(+1,+1)$

- Let Y_i denote the four-component row vector of potential outcomes for unit i : $Y_i = [Y_i(-1, -1), Y_i(-1, +1), Y_i(+1, -1), Y_i(+1, +1)]$.
- Let us introduce $g_{F_1} = (-1, -1, +1, +1)^T$ and $g_{F_2} = (-1, +1, -1, +1)^T$.

Lexicographic order of treatment combinations : example with two factors, each having two levels

- Let Y_i denote the four-component row vector of potential outcomes for unit i : $Y_i = [Y_i(-1, -1), Y_i(-1, +1), Y_i(+1, -1), Y_i(+1, +1)]$.
- Let us introduce $g_{F_1} = (-1, -1, +1, +1)^T$ and $g_{F_2} = (-1, +1, -1, +1)^T$.
- Write $\tau_{i, F_1} = \frac{1}{2}[Y_i(+1, -1) - Y_i(-1, -1) + Y_i(+1, +1) - Y_i(-1, +1)]$ as a function of g_{F_1} and Y_i .

Lexicographic order of treatment combinations : example with two factors, each having two levels

- Let Y_i denote the four-component row vector of potential outcomes for unit i : $Y_i = [Y_i(-1, -1), Y_i(-1, +1), Y_i(+1, -1), Y_i(+1, +1)]$.
- Let us introduce $g_{F_1} = (-1, -1, +1, +1)^T$ and $g_{F_2} = (-1, +1, -1, +1)^T$.
- Write $\tau_{i, F_2} = \frac{1}{2}[Y_i(-1, +1) - Y_i(-1, -1) + Y_i(+1, +1) - Y_i(+1, -1)]$ as a function of g_{F_2} and Y_i .

Lexicographic order of treatment combinations : example with two factors, each having two levels

Treatment	F ₁	F ₂	Unit POs
(-1,-1)	-1	-1	$Y_i(-1,-1)$
(-1,+1)	-1	+1	$Y_i(-1,+1)$
(+1,-1)	+1	-1	$Y_i(+1,-1)$
(+1,+1)	+1	+1	$Y_i(+1,+1)$

- Write $\tau_{i,F_1*F_2} = \frac{1}{2} \{ [Y_i(+1, +1) - Y_i(-1, +1)] - [Y_i(+1, -1) - Y_i(-1, -1)] \}$ as a function of $g_{F_1*F_2}$ and Y_i .

Example of a 2^3 randomized experiment

- Three factors, each having two levels.

Treatment	F_1	F_2	F_3	Unit POs
$(-1,-1,-1)$				$Y_i(-1,-1,-1)$
$(-1,-1,+1)$				$Y_i(-1,-1,+1)$
$(-1,+1,-1)$				$Y_i(-1,+1,-1)$
$(-1,+1,+1)$				$Y_i(-1,+1,+1)$
$(+1,-1,-1)$				$Y_i(+1,-1,-1)$
$(+1,-1,+1)$				$Y_i(+1,-1,+1)$
$(+1,+1,-1)$				$Y_i(+1,+1,-1)$
$(+1,+1,+1)$				$Y_i(+1,+1,+1)$

Assignment mechanism

- $P(W_i=(-1,-1,-1))=\frac{N_{(-1,-1,-1)}}{N}$ [and so on..]

- Assignment mechanism :

$$P(\mathbf{W}|\mathbf{X},\mathbf{Y}(0),\mathbf{Y}(1)) = \begin{cases} \binom{N}{N_{(-1,-1,-1)}}^{-1} \dots \binom{N-N_{(-1,-1,-1)}-\dots-N_{(+1,-1,+1)}}{N_{(+1,+1,-1)}}^{-1} \\ \text{for } \mathbf{W} \in \mathbb{W}^+ \\ 0 \text{ otherwise} \end{cases}$$

where

$$\mathbb{W}^+ = \left\{ \mathbf{W} \text{ such that } \sum_{i=1}^N \mathbb{1}[W_i = (-1, -1, -1)] = N_{(-1,-1,-1)} \text{ [and so on..]} \right\}$$

Science Table

- Let Y_i denote the eight component row vector of potential outcomes for unit i :

$$Y_i = [Y_i(-1, -1, -1), Y_i(-1, -1, +1), Y_i(-1, +1, -1), Y_i(-1, +1, +1), Y_i(+1, -1, -1), Y_i(+1, -1, +1), Y_i(+1, +1, -1), Y_i(+1, +1, +1)]$$

Unit i	$Y_i(-1, -1, -1)$	$Y_i(-1, -1, +1)$	...	$Y_i(+1, +1, +1)$
1	$Y_1(-1, -1, -1)$	$Y_1(-1, -1, +1)$	...	$Y_1(+1, +1, +1)$
2	$Y_2(-1, -1, -1)$	$Y_2(-1, -1, +1)$	...	$Y_2(+1, +1, +1)$
...	...	...	...	...
...	...	...	...	...
N	$Y_N(-1, -1, -1)$	$Y_N(-1, -1, +1)$	...	$Y_N(+1, +1, +1)$

- Causal estimands

Science Table

- Let Y_i denote the eight component row vector of potential outcomes for unit i :

$$Y_i = [Y_i(-1, -1, -1), Y_i(-1, -1, +1), Y_i(-1, +1, -1), Y_i(-1, +1, +1), Y_i(+1, -1, -1), Y_i(+1, -1, +1), Y_i(+1, +1, -1), Y_i(+1, +1, +1)]$$

- Causal estimands

- Unit-level causal effects of factor 1 :

$$\tau_{i,F_1} = \frac{1}{4} Y_i g_{F_1}$$

- Unit-level causal effects of factor 2 :

$$\tau_{i,F_2} = \frac{1}{4} Y_i g_{F_2}$$

- Unit-level causal effects of factor 3 :

$$\tau_{i,F_3} = \frac{1}{4} Y_i g_{F_3}$$

- Causal estimands

- Unit-level two-factor interactions between F_1 and F_2 :

$$\tau_{i,F_1*F_2} = \frac{1}{4} Y_i g_{F_1*F_2}$$

- Unit-level two-factor interactions between F_1 and F_3 :

$$\tau_{i,F_1*F_3} = \frac{1}{4} Y_i g_{F_1*F_3}$$

- Unit-level two-factor interactions between F_2 and F_3 :

$$\tau_{i,F_2*F_3} = \frac{1}{4} Y_i g_{F_2*F_3}$$

- Causal estimands

- Unit-level three-factor interactions between F_1 , F_2 , and F_3 :

$$\tau_{i,F_1*F_2*F_3} = \frac{1}{4} Y_i - g_{F_1*F_2*F_3}$$

Generalization to 2^K

- Let us consider a balanced 2^K factorial design with N units.
- The unit-level causal effects for a general 2^K randomized experiment can be defined as :

$$\tau_{i,F} = \frac{1}{2^{K-1}} Y_i g_F$$

where $F \in \{F_1, F_2, \dots, F_K, F_1 * F_2, \dots, F_{K-1} * F_K, \dots, F_1 * F_2 * \dots * F_K\}$.

- The average causal effects are defined as :

$$\tau_F = \frac{1}{N} \sum_{i=1}^N \tau_{i,F}$$

Observed Table

Unit i	W_i	$Y_i(-1, -1, -1)$	...	$Y_i(+1, +1, +1)$
1	$(-1, -1, -1)$	$Y_1(-1, -1, -1)$	...	?
2	$(+1, +1, +1)$	?	...	$Y_2(+1, +1, +1)$
...	...	...	...	...
...	...	...	...	...
N	$(-1, +1, -1)$	?	...	?

$$Y_i^{obs} = \sum_w \mathbb{1}(W_i = w) Y_i(W_i = w)$$

Fisherian inference for a 2^K design

Unit i	W_i	$Y_i(-1, \dots, -1)$	...	$Y_i(+1, \dots, +1)$
1	$(-1, \dots, -1)$	$Y_1(-1, \dots, -1)$	...	?
2	$(+1, \dots, +1)$	?	...	$Y_2(+1, \dots, +1)$
...	...	...	...	...
...	...	...	...	...
N	$(-1, \dots, +1)$	?	...	?

- Write down the sharp null hypothesis for 2^K :

Fisherian inference for a 2^K design

Unit i	W_i	$Y_i(-1, \dots, -1)$	...	$Y_i(+1, \dots, +1)$
1	$(-1, \dots, -1)$	$Y_1(-1, \dots, -1)$	...	?
2	$(+1, \dots, +1)$	?	...	$Y_2(+1, \dots, +1)$
...	...	...	...	...
...	...	...	...	...
N	$(-1, \dots, +1)$	?	...	?

- Test statistic :

Neymanian inference

Unit i	W_i	$Y_i(-1, \dots, -1)$	...	$Y_i(+1, \dots, +1)$
1	$(-1, \dots, -1)$	$Y_1(-1, \dots, -1)$	...	?
2	$(+1, \dots, +1)$	?	...	$Y_2(+1, \dots, +1)$
...	...	...	...	...
...	...	...	...	...
N	$(-1, \dots, +1)$	?	...	?

- Estimator for τ_F ?

Neymanian inference

- Estimator for the sampling variance of $\hat{\tau}_F$:

- $\hat{Var}(\hat{\tau}_F) = \frac{2^K}{N} \hat{S}^2$

where $\hat{S}^2 = \frac{1}{2^K} [s_{(-1, \dots, -1)}^2 + \dots + s_{(+1, \dots, +1)}^2]$

e.g., $s_{(-1, \dots, -1)}^2 = \frac{1}{\frac{N}{2^K} - 1} \sum_{i: W_i = (-1, \dots, -1)} [Y_i^{obs} - \bar{Y}_{(-1, \dots, -1)}^{obs}]^2$

[Unbiased assuming $S_{(-1, \dots, -1)}^2 = \dots = S_{(+1, \dots, +1)}^2 = S^2$ and $S_F^2 = 0$]